

CAP182 SS2

PART A

Title	Effective hospital readmission prediction models using machine-learned features.
Harvard reference	Davis, S., Zhang, J., Lee, I., Rezaei, M., Greiner, R., McAlister, F.A. and Padwal, R. (2022) 'Effective hospital readmission prediction models using machine-learned features', BMC Health Services Research, 22, article 1415. Available at: https://doi.org/10.1186/s12913-022-08748-y (Accessed: 25 September 2026).
Link	https://doi.org/10.1186/s12913-022-08748-y
Modelling	Logistic Regression and Gradient Boosting Machine (GBM) models were compared. A tuned GBM used a learning rate of 0.01, maximum tree depth of 8 and 1,000 estimators. The researchers also evaluated the LACE score as a baseline. Ten-fold cross-validation and a separate holdout test set were used. Word2Vec was used to create features from longitudinal patient records.
Preprocessing techniques	The researchers selected an index hospital admission for each patient, excluded specified records such as deaths, transfers and psychiatric admissions, and linked hospital, outpatient, physician, prescription and laboratory records. They engineered features from previous healthcare utilisation, comorbidities, prescriptions and diagnoses. Word2Vec converted chronological medical codes into numerical representations. The final patient representation used the sum of the last 25 medical-code vectors.
Datasets used	Linked administrative healthcare records from Alberta, Canada, covering 2011–2017. Sources included the Discharge Abstract Database, National Ambulatory Care Reporting System, physician claims, Pharmaceutical Information Network and laboratory records. The study also considered Canadian Census information for selected social and economic variables, which were removed because they did not improve performance.
Key takeaways	1. Compare a simple, interpretable baseline such as Logistic Regression with a tree-based ensemble model. 2. Previous admissions, length of stay, diagnoses and healthcare utilisation can be useful predictors of readmission risk. 3. It is important to use a separate test set and cross-validation to evaluate whether the model generalises to unseen data.

Title	Leveraging a machine learning model to predict hospital readmission risk: integrating clinical and social determinants of health data.
Harvard reference	Zhang, T. (2026) 'Leveraging a machine learning model to predict hospital readmission risk: integrating clinical and social determinants of health data', <i>Frontiers in Public Health</i> , 14, article 1754585. Available at: https://doi.org/10.3389/fpubh.2026.1754585 (Accessed: 25 September 2026).
Link	https://doi.org/10.3389/fpubh.2026.1754585
Modelling	Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost and Support Vector Machine were compared. Bayesian optimisation and five-fold cross-validation were used for hyperparameter tuning. The data were divided into training (70%), validation (15%) and test (15%) sets. SHAP and the Anchor algorithm were used to interpret model predictions.
Preprocessing techniques	The study applied inclusion and exclusion criteria to construct the cohort, removed records with missing essential clinical or demographic information, and examined missingness in the remaining variables. Patient addresses were geocoded and linked to census tract-level social indicators. Stratified data splitting and geographic grouping during cross-validation were used to support model evaluation and reduce information leakage
Datasets used	Electronic health records from a large academic medical centre covering January 2022 to December 2023, combined with publicly available American Community Survey 2018–2022 census estimates. The final dataset contained 3,018 patients. Clinical variables included demographics, previous healthcare utilisation, comorbidity burden and hospital encounter characteristics.
Key takeaways	1. Compare different classification algorithms rather than assuming one model will perform best. 2. Evaluate performance using ROC-AUC, precision-recall AUC, F1-score, sensitivity and specificity, particularly when the readmission outcome is imbalanced. 3. The use of model interpretation techniques such as SHAP to identify influential predictors and support understanding of model outputs.

Title	Predicting 30-Day Hospital Readmission in Patients With Diabetes Using Machine Learning on Electronic Health Record Data.
Harvard reference	Emi-Johnson, A. et al. (2025) 'Predicting 30-Day Hospital Readmission in Patients With Diabetes Using Machine Learning on Electronic Health Record Data'. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC12085305/
Link	https://pmc.ncbi.nlm.nih.gov/articles/PMC12085305/
Modelling	The study compared four models: Logistic Regression, Random Forest, XGBoost and Deep Neural Networks (DNN). SHAP (SHapley Additive exPlanations) was used to identify influential predictors and interpret model outputs. The models were evaluated using an 80/20 stratified train-test split. XGBoost achieved the highest reported ROC-AUC of 0.667, followed by Logistic Regression at 0.642 and Random Forest at 0.630.
Preprocessing techniques	Missing-value imputation, feature scaling and one-hot encoding were applied. Records with missing or ambiguous readmission labels were removed. The dataset was divided using stratified sampling to preserve the readmission class distribution.
Datasets used	Diabetes 130-US Hospitals for Years 1999–2008, obtained from the UCI Machine Learning Repository. The dataset contains 101,766 de-identified inpatient encounters from 130 US hospitals, with demographic, diagnostic, medication, admission and discharge information.
Key takeaways	1. Compare several models, including Logistic Regression, Random Forest and XGBoost, rather than assuming a complex model will perform best. 2. Apply preprocessing carefully, including missing-value handling, categorical encoding and scaling where appropriate. 3. Use recall, precision, F1-score and ROC-AUC alongside accuracy.

Name	Diabetes 130-US Hospitals for Years 1999–2008
Harvard reference	Strack, B., DeShazo, J.P., Gennings, C., Olmo, J.L., Ventura, S., Cios, K.J. and Clore, J.N. (2014) 'Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records', BioMed Research International, 2014, Article ID 781670. Available at: https://doi.org/10.1155/2014/781670 .
Link	https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
Number of instances	101,766 hospital encounter records (rows). The dataset does not have predefined training, validation or testing splits. I will divide the data into training and testing sets during preprocessing, ensuring that encounters belonging to the same patient do not appear in both sets.
Number of feature columns	50 original columns, including the target variable. These include patient demographics, admission details, diagnoses, medications and hospital encounter information. The final number of input features will be determined after preprocessing and feature selection.
Number of classes (if applicable)	2 classes for the binary classification task. Class 1 represents readmission within 30 days (<30), while Class 0 represents no recorded readmission within 30 days (NO or >30).
Target range (if applicable)	Not applicable. This is a classification task rather than a regression task.
Origin	The dataset is publicly available through the UCI Machine Learning Repository. It contains de-identified hospital encounter records from 130 US hospitals collected between 1999 and 2008. The data includes patient demographics, diagnoses, medication information, admission details and readmission outcomes. It is historical US hospital data and does not represent current South African hospital records.